PROBLEM K10

Planar Flow with Exponentially Varying Shear (Quiz, 30 September 2013)

Original statement · K10.pdf

Unsteady | Potential | Stream function | Pure shear

OFFICIAL PROBLEM STATEMENT

Given the steady planar motion described by the velocity components

$$v_x = Ae^{\Omega t}\, y \qquad \text{and} \qquad v_y = Be^{\Omega t}\, x,$$

where A, B, and Ω are constants.

1. Determine the value of the volume dilation rate $\nabla \cdot \vec{v}$.
2. Obtain the value of B for which the resulting flow is irrotational. Adopt this particular value in what follows.
3. Calculate the stream line that intersects the point $(x, y) = (x_0, y_0)$ at a given instant of time t. Sketch the resulting stream lines.
4. Compute the trajectory of a fluid particle located initially at $(x, y) = (x_0, y_0)$.
5. If they exist, calculate the velocity potential φ and the stream function ψ.
6. Determine the rate-of-strain tensor $\bar{\bar{T}}_d$.
7. Describe the deformation of a fluid element of square shape whose sides of length dl are initially oriented parallel to the axes.

Source: K10.pdf (page 1). Notes.pdf, Chapter 2: expansion rate Eqs. (2.62)-(2.63); vorticity Eq. (2.52); stream lines Eq. (2.21); trajectories Eq. (2.12); potential Eq. (2.34); stream function Eqs. (2.40)-(2.45); velocity gradient and rate-of-strain tensor Eqs. (2.46)-(2.48); square element Eqs. (2.56)-(2.61).

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar flow with a velocity that is linear in the coordinates and multiplied by the common time factor $e^{\Omega t}$. Two spatial degrees of freedom and an explicit time dependence.
- Although the statement calls the motion "steady", the velocity depends explicitly on t whenever $\Omega \neq 0$; the flow is unsteady in general (and part 3 asks for the streamline "at a given instant of time t"). The steady case is recovered for $\Omega = 0$. The general case $\Omega \neq 0$ is solved below, with $A \neq 0$.
- The constants A, B and Ω have units s^{-1} (so that $v = Ae^{\Omega t}y$ is a velocity).

Symbol	Meaning	SI unit
A, B	shear constants	s^{-1}
Ω	rate of the exponential time dependence	s^{-1}
(x_0, y_0)	position at the reference (initial) instant	m
$\tau(t) = (A/\Omega)\left(e^{\Omega t} - 1\right)$	dimensionless time variable defined in 3.4	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y)$, origin at the stagnation point $\mathbf{v} = 0$. Gradient convention $(\nabla \mathbf{v})_{ij} = \partial v_j / \partial x_i$.
- Change of time variable for part 4: $\tau(t) = \dfrac{A}{\Omega}\left(e^{\Omega t} - 1\right)$, with $d\tau = Ae^{\Omega t}dt$ and $\tau(0) = 0$. In terms of τ the unsteady trajectory system becomes autonomous with constant coefficients.
- Change of basis for the sum and difference coordinates $u = x + y$, $w = x - y$ (a rotation by 45° and a scaling by $\sqrt{2}$) used to decouple the trajectory equations; this is the same basis that diagonalises $\bar{\bar{T}}_d$ in part 6.

PHASE 3

Step-by-Step Derivation

3.1 Volume dilation rate

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate at which a material volume changes per unit volume (Eqs. 2.62-2.63).

$$\nabla \cdot \mathbf{v} = \frac{\partial}{\partial x}\left(Ae^{\Omega t}y\right) + \frac{\partial}{\partial y}\left(Be^{\Omega t}x\right) = 0 + 0 = \boxed{0}$$

for any values of A, B and Ω: the flow is incompressible.

3.2 Condition for irrotational flow

Why this tool: the flow is irrotational when $\boldsymbol{\omega} = \nabla \wedge \mathbf{v} = 0$ (text after Eq. 2.33).

$$\omega_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = Be^{\Omega t} - Ae^{\Omega t} = (B - A)e^{\Omega t}.$$

Since $e^{\Omega t} > 0, \omega_z = 0$ for all t if and only if

$$\boxed{B = A.}$$

From now on $\mathbf{v} = Ae^{\Omega t}(y, x)$.

3.3 Streamlines at a given instant

Why this tool: streamlines are tangent to the velocity at a frozen instant (Eq. 2.21); t is a parameter.

$$\frac{dx}{v_x} = \frac{dy}{v_y} \implies \frac{dx}{Ae^{\Omega t}y} = \frac{dy}{Ae^{\Omega t}x} \implies \frac{dx}{y} = \frac{dy}{x} \implies x\,dx = y\,dy,$$

where the common non-zero factor $Ae^{\Omega t}$ cancels. Integrate between (x_0, y_0) and (x, y) (Barrow's rule):

$$\left[\frac{x'^2}{2}\right]_{x_0}^{x} = \left[\frac{y'^2}{2}\right]_{y_0}^{y} \implies \boxed{y^2 - x^2 = y_0^2 - x_0^2.}$$

These are rectangular hyperbolae with asymptotes $y = \pm x$; the shape does not depend on t (only the speed does). For $y_0^2 > x_0^2$ the branches open towards $\pm y$, for $y_0^2 < x_0^2$ towards $\pm x$, and $y_0^2 = x_0^2$ gives the straight lines $y = \pm x$. Sketch (in words), for $Ae^{\Omega t} > 0$: v_x has the sign of y and v_y the sign of x, so in the first and third quadrants the flow moves away from the origin along the diagonal $y = x$, and in the second and fourth quadrants towards the origin along $y = -x$. The origin is a stagnation point; it is a saddle, with $y = -x$ the incoming and $y = x$ the outgoing direction. For $Ae^{\Omega t} < 0$ all arrows reverse.

3.4 Trajectory of the particle starting at (x_0, y_0)

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}(\mathbf{x}, t)$ with $\mathbf{x}(0) = (x_0, y_0)$ (Eq. 2.12). Because $\mathbf{v}$ is a separable product of a function of time and a function of position, the change of time variable τ of Phase 2 makes the system autonomous. With $d\tau = Ae^{\Omega t}dt$:

$$\frac{dx}{d\tau} = y, \qquad \frac{dy}{d\tau} = x.$$

Sum and difference: $u = x + y$ and $w = x - y$ satisfy

$$\frac{du}{d\tau} = y + x = u, \qquad \frac{dw}{d\tau} = y - x = -w.$$

Integrate with Barrow's rule between $\tau = 0$ and τ: $\ln(u/u_0) = \tau$ and $\ln(w/w_0) = -\tau$, i.e. $u = (x_0 + y_0)e^{\tau}$ and $w = (x_0 - y_0)e^{-\tau}$. Return to $x = (u + w)/2$, $y = (u - w)/2$:

$$\boxed{x = x_0 \cosh\tau + y_0 \sinh\tau, \qquad y = y_0 \cosh\tau + x_0 \sinh\tau, \qquad \tau = \frac{A}{\Omega}\left(e^{\Omega t} - 1\right).}$$

Path lines: eliminate τ using $\cosh^2\tau - \sinh^2\tau = 1$:

$$y^2 - x^2 = (y_0^2 - x_0^2)\left(\cosh^2\tau - \sinh^2\tau\right) = y_0^2 - x_0^2,$$

the same hyperbolae as the streamlines. (The path lines coincide with the streamlines because the direction of $\mathbf{v}$ does not depend on time.)

Logarithmic form (equivalent, as in the handwritten solution). Along the path $y = \pm\sqrt{x^2 + y_0^2 - x_0^2}$, so $dx/dt = \pm A e^{\Omega t}\sqrt{x^2 + C}$ with $C = y_0^2 - x_0^2$. Since $\dfrac{d}{dx}\ln\left(x + \sqrt{x^2 + C}\right) = \dfrac{1}{\sqrt{x^2 + C}}$, Barrow's rule gives, for $y_0 > 0$ (upper sign),

$$\ln\frac{x + \sqrt{x^2 + y_0^2 - x_0^2}}{x_0 + y_0} = \frac{A}{\Omega}\left(e^{\Omega t} - 1\right),$$

which is $\ln[(x+y)/(x_0+y_0)] = \tau$ obtained from $u = u_0 e^{\tau}$. For $y_0 < 0$ the lower branch of the square root applies.

3.5 Velocity potential and stream function

Why this tool: the flow is irrotational (3.2) so a potential exists (Eq. 2.34); it is planar and solenoidal (3.1) so a stream function exists (Eq. 2.41).

Potential: $\partial_x\varphi = v_x = A e^{\Omega t} y$ gives $\varphi = A e^{\Omega t} xy + f(y,t)$; $\partial_y\varphi = A e^{\Omega t} x + f' = v_y = A e^{\Omega t} x$ gives $f' = 0$:

$$\boxed{\varphi = A e^{\Omega t}\, xy + \varphi_0(t).}$$

Stream function: $\partial_y\psi = v_x = A e^{\Omega t} y$ gives $\psi = \frac{1}{2} A e^{\Omega t} y^2 + g(x,t)$; $-\partial_x\psi = -g' = v_y = A e^{\Omega t} x$ gives $g = -\frac{1}{2} A e^{\Omega t} x^2$:

$$\boxed{\psi = \frac{A e^{\Omega t}}{2}\left(y^2 - x^2\right) + \psi_0(t).}$$

Check: $\partial_y\psi = A e^{\Omega t} y = v_x$ and $-\partial_x\psi = A e^{\Omega t} x = v_y$. The level curves $\psi = \text{const}$ are the hyperbolae of 3.3 (Eq. 2.42), and are orthogonal to the level curves of φ ($xy = \text{const}$).

Reading of the official handwritten solution

K10.pdf (part 5) gives $\psi = Ae^{\Omega t}(y^2 - x^2) + \psi_0$, without the factor $1/2$. That expression would give $\partial_y \psi = 2Ae^{\Omega t} y$, twice the velocity component v_x. The relations written just above that line, $\partial_y \psi = Ae^{\Omega t} y$ and $\partial_x \psi = -Ae^{\Omega t} x$, are correct, and integrating them produces the factor $1/2$. The correct result is $\psi = \frac{1}{2} Ae^{\Omega t}(y^2 - x^2) + \psi_0$. The statement also calls the motion "steady", although the velocity depends on t through $e^{\Omega t}$ (see Phase 1). The velocity potential $\varphi = Ae^{\Omega t} xy + \varphi_0$ in the handwritten solution is correct.

3.6 Rate-of-strain tensor

Why this tool: $\bar{\bar{T}}_d = \frac{1}{2}\left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T\right]$ (Eq. 2.47) gives the pure strain of the material elements.

$$\nabla \mathbf{v} = \begin{pmatrix} \partial_x v_x & \partial_x v_y \\ \partial_y v_x & \partial_y v_y \end{pmatrix} = Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \boxed{\bar{\bar{T}}_d = Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \nabla \mathbf{v}.}$$

It is symmetric, so $\bar{\bar{T}}_r = \frac{1}{2}\left[\nabla \mathbf{v} - (\nabla \mathbf{v})^T\right] = 0$, consistent with $\omega_z = 0$. Its eigenvalues follow from $\lambda^2 - (Ae^{\Omega t})^2 = 0$: $\lambda = \pm Ae^{\Omega t}$, with principal directions $(1, 1)/\sqrt{2}$ (extension, for $Ae^{\Omega t} > 0$) and $(1, -1)/\sqrt{2}$ (compression); these are the outgoing and incoming diagonals found in 3.3. The trace is zero ($\nabla \cdot \mathbf{v} = 0$).

3.7 Deformation of a square fluid element

Why this tool: a material segment evolves as $d\mathbf{l}(t + dt) = d\mathbf{l} + (d\mathbf{l} \cdot \nabla \mathbf{v})\, dt$ (Eqs. 2.46 and 2.56-2.57). With the sides along the axes:

$$dl\,(1,0) \;\to\; dl\,(1,0) + dl\,(1,0)\, Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} dt = dl\,\left(1,\; Ae^{\Omega t} dt\right),$$

$$dl\,(0,1) \;\to\; dl\,(0,1) + dl\,(0,1)\, Ae^{\Omega t} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} dt = dl\,\left(Ae^{\Omega t} dt,\; 1\right).$$

The horizontal side tilts upward by the angle $Ae^{\Omega t} dt$ and the vertical side tilts towards $+x$ by the same angle, so the two sides rotate towards each other, and their (initially right) angle decreases by $2Ae^{\Omega t} dt$. There is no net rotation ($\bar{\bar{T}}_r = 0$): the square deforms into a

rhombus-like parallelogram by pure shear strain, with strain rate $\dot{\gamma} = \partial_x v_y + \partial_y v_x = 2Ae^{\Omega t}$ (the rate at which the angle between the sides decreases, Notes after Eq. 2.61). The area changes only at second order ($dl^2\,\left[1 - (Ae^{\Omega t}dt)^2\right]$), as required by $\nabla \cdot \mathbf{v} = 0$. Equivalently, the diagonal along $y = x$ is stretched at the rate $Ae^{\Omega t}$ and the diagonal along $y = -x$ is compressed at the same rate.

PHASE 4

Interpretation, Limits and Dimensional Check

- Physical picture: it is the stagnation-point flow of K7 rotated by 45° (principal axes along the diagonals, no rotation), with a strain rate $Ae^{\Omega t}$ that grows or decays exponentially in time.
- Steady limit $\Omega \to 0$: $\tau \to At$, and the trajectories become $x = x_0 \cosh At + y_0 \sinh At$, $y = y_0 \cosh At + x_0 \sinh At$, the standard stagnation-point trajectories. For $\Omega < 0$ the time variable saturates at $\tau_\infty = A/|\Omega|$, so particles approach finite limit positions (the strain rate dies out) instead of escaping.
- Particles on the incoming diagonal $y_0 = -x_0$ satisfy $u = 0$ for all t and $w = w_0 e^{-\tau}$: they move towards the origin as τ grows, but reach it only as $\tau \to \infty$ (for $\Omega > 0$, $A > 0$, as $t \to \infty$).
- Consistency: $\bar{\bar{T}}_d$ symmetric traceless, $\omega_z = 0$, $\nabla \cdot \mathbf{v} = 0$; $\psi =$ const reproduces the streamlines of 3.3 and the paths of 3.4.
- Dimensions (SI): $[Ae^{\Omega t}y] = \mathrm{s}^{-1}\mathrm{m} = \mathrm{m/s}$; $[\tau] = [A/\Omega] = 1$; $[\varphi] = [Axy] = \mathrm{m}^2/\mathrm{s}$; $[\psi] = [Ay^2] = \mathrm{m}^2/\mathrm{s}$; $[\bar{\bar{T}}_d] = [A] = \mathrm{s}^{-1}$; $[\dot{\gamma}] = \mathrm{s}^{-1}$.
- Numerical cross-check: RK4 integration of the particle ODE with $A = B = 0.8\ \mathrm{s}^{-1}$, $\Omega = 0.9\ \mathrm{s}^{-1}$, $(x_0, y_0) = (0.5, 1.2)\ \mathrm{m}$ up to $t = 1.7\ \mathrm{s}$ gives $(21.17827, 21.20635)\ \mathrm{m}$, equal to the cosh/sinh formulas ($\tau = 3.216157$), with $y^2 - x^2 = 1.19\ \mathrm{m}^2$ equal to $y_0^2 - x_0^2$, and $\ln[(x+y)/(x_0+y_0)] = 3.216157 = \tau$.